<u>**Technical Report**</u>

Team: Insightfy

Data	Adventures works
Tools	Excel-SQL-Python-Power BI-power query

Name	ID
Youssef Sayed Tolba	21073546
Mariam abdelbaki mohamed	21085866
Eyad Muhammad Abdel Azim	21124890
Mostafa Hossam Ismail	21065952
Mostafa Mohamed Shaban	21091221
Youssef taher bakry	21082950

Project Description

Project Description This project analyzes the performance of sales operations and supply chain activities using the AdventureWorks dataset. The objective is to evaluate how product demand, inventory levels, and supplier performance impact overall sales efficiency and business operations. The analysis focuses on identifying sales trends, top-performing products, customer purchasing patterns, and potential supply chain inefficiencies such as low inventory levels or delayed shipments. By integrating sales data with product and inventory information, the project aims to provide insights that support better demand planning, inventory optimization, and strategic decision-making. The final deliverables include an interactive dashboard that highlights key business metrics, enabling stakeholders to monitor performance and identify opportunities for improving operational efficiency and profitability.

1. Project Summary & Problem Statement

For any successful business, it is vital to balance selling products quickly with keeping the supply chain running smoothly. This report documents a data analysis project built using the **AdventureWorks** company dataset. The main goal of this project is to find business problems where product demand, changing stock levels, and supplier issues create delays and lower company profits.

By gathering data from different parts of the business, our analysis uncovers deep patterns in sales orders, revenue tracking, supplier lead times, and stock counts. We organized this information into a clean Star Schema database model and displayed it in a Power BI dashboard. This final interactive system helps company managers make faster, smarter decisions about inventory levels, purchasing, and sales plans.

2. Data Sources Overview

Our analysis is based on the **AdventureWorks2025** database. To set up our dashboards, transaction records were extracted from five main business areas:

- **Sales Area:** Contains metrics about customer profiles, store locations, regional sales areas, sales team performance quotas, credit card records, and specific promotions.
- **Production Area:** Tracks product details, categories, subcategories, warehouse site locations, and real-time physical stock counts.
- **Purchasing Area:** Documents vendor profiles, vendor credit ratings, shipping methods, historical purchase order statuses, and product rejection reasons.
- **Person Area:** Houses fundamental master contact information, helping us separate individual shoppers from business clients.
- **Human Resources (HR) Area:** Maps company department structures, providing layout definitions for our internal team members and regional sales representatives.

3. Technical Architecture Steps

The project followed a clean, logical 4-step data pipeline to turn raw data into actionable dashboard insights:

1. **Step 1 - SQL Extraction:** We wrote optimized SQL scripts to pull data from the relational database. This organized the records into transactional event tables and descriptive lookups.
2. **Step 2 - Python Cleaning:** We processed the extracted data files in Python. Using the Pandas library, we performed data profiling, filled in empty fields, and grouped categories.
3. **Step 3 - Power Query Transformations:** We loaded the clean datasets into Power Query to format them uniformly. This included combining name fields and sorting geographic areas.
4. **Step 4 - Power BI Visualizations:** We organized the final data into a high-performance Star Schema model inside Power BI and built interactive dashboards for business reviews.

4. Preparing Data with SQL

We used custom SQL scripts to split database information into Fact tables (transaction details like sales and purchases) and Dimension tables (descriptive details like customer or product lookups). A major optimization step involved distributing header-level operational costs (like tax and freight totals) down to individual item lines. This allocation was calculated proportionally across order rows while using NULLIF functions to prevent mathematical divide-by-zero errors on cancelled transactions.

Fact Table Construction Strategies

The following data tables were extracted via SQL queries to build the core foundation of our analytical schema:

- **F_Sales Table:** Merges sales order header and detail lines together. It creates numeric date tracking keys formatted as text strings (YYYYMMDD) and allocates costs across line items with the following logic:
  `ROUND(TaxAmt * (LineTotal / NULLIF(OrderTotal, 0)), 3) AS [Allocated Tax]`
- **F_Purchasing Table:** Connects purchasing items with order parameters. It tracks inventory counts such as ordered, received, stocked, and rejected quantities across specific procurement dates.
- **F_Transaction_History Table:** Consolidates older archive history data and live transactional database records using a clear UNION ALL clause. It groups transactional categories into clear descriptions ('Work Order', 'Sales Order', 'Purchase Order') and determines direction based on product volumes:

CASE WHEN Quantity > 0 THEN 'Stock In' ELSE 'Stock Out' END

- **F_Inventory_Snapshot Table:** Creates unified warehouse location codes by combining shelf rows and storage bin details into a single tracking string:
  CAST(pi.LocationID AS VARCHAR(10)) + '-' + ISNULL(pi.Shelf, 'N/A') + '-' + CAST(pi.Bin AS VARCHAR(10))

Dimension Table Setup

- **Dim_Product**
  - Stores all descriptive information for your items.
  - It contains product names, product lines, models, category names, subcategory names, list prices, and colors.
  - This table helps identify which specific products drive the highest customer demand.

- **Dim_Customer**
  - Houses master customer profiles cleaned using Python data scripts.
  - It tracks the customer's Full Name, associated retail store name, and active territory keys.
  - It features the engineered Customer Type flag to easily separate individual consumers (B2C) from wholesale business stores (B2B).

- **Dim_Vendor**
  - Stores data about the third-party companies supplying your raw materials and inventory items.
  - It records vendor names, active status flags, and strategic preferred partner tags.
  - It includes vendor credit ratings to ensure suppliers are meeting corporate financial reliability standards.

- **Dim_SalesPerson**
  - Records details about your internal enterprise sales team members.
  - It fields employee names, designated sales territories, and historical commission percentages.
  - It tracks individual performance metrics including specific quotas, annual bonuses, and year-to-date sales achievements.

- **Dim_Territory**
  - Organizes the company's regional markets into clear geographical boundaries.

- It maps specific territory names, localized country codes, and macro-level territory groups (like North America or Europe).
- This allows management to easily compare annual sales volumes and profits across different global regions.

- **Dim_location**

  - Tracks the physical manufacturing, assembly, or production workstations within the factory floors.
  - It contains specific location names, baseline operational cost rates, and workspace capacity availability percentages.

- **Dim_InventoryLocation**

  - Pinpoints exactly where products are stored inside the physical warehouse storage facilities to make inventory tracking easy.
  - It organizes warehouse storage spots by combining location IDs, shelf rows, and precise tracking bin numbers.

- **Dim_ShipMethod**

  - Details the transport logistics and configurations used for shipping products to customers or receiving items from vendors.
  - It documents official shipping company names, baseline handling costs, and cost factors based on item weight scales.

- **Dim_Special_Offer**

  - Manages sales promotional campaigns, marketing events, and active discount structures.
  - It monitors offer descriptions, specific discount percentages, and active start or end date ranges.
  - It sets minimum and maximum purchase quantity rules required for the discount to automatically apply.

- **Dim_CreditCard**

  - A basic dimensional reference table that categorizes the card payment type models used by customers during checkouts.
  - It separates card transaction records into types like Visa or MasterCard to analyze preferred payment choices.

- **Dim_SalesOrderStatus**

- Turns raw numerical database status flags into clear corporate phrases for sales order fulfillment tracking.
- It maps order states into human-readable labels such as In Process, Approved, Backordered, Rejected, Shipped, or Cancelled.

- **Dim_PurchaseOrderStatus**
    - Translates system status flags for vendor procurement orders into clear, non-technical tracking steps.
    - It categorizes the lifecycle of supplier purchases into four states: Pending, Approved, Rejected, or Completed.

5. Cleaning Data with Python

To ensure our analytical dashboards display accurate information, we used Python and the Pandas library to clean the extracted data files, with a special focus on the DimCustomer data matrix.

Sample Data Review

Running initial data scripts on the customer file showed a total of 19,820 records distributed across 7 columns. Checking for missing fields revealed several gaps in the data matrix:

Customer Column	Missing Values Found	Data Cleaning Analysis Conclusion
Customer ID	0	Fully populated customer primary key index column.
Territory ID	0	Fully populated sales tracking region classification field.
First Name / Last Name	701	Missing fields match rows for wholesale commercial business stores.
Middle Name	8,800	Highly sparse column; not essential for our sales

Customer Column	Missing Values Found	Data Cleaning Analysis Conclusion
		tracking analysis.
Person Type	**701**	Missing records match raw database data gaps.
Store Name	**18,484**	Naturally empty for direct individual retail customers.

Data Wrangling Tasks Completed

We resolved these data gaps systematically using specialized Python data operations:

- **Dropping Sparse Columns:** Since Middle Name was blank in 44.3% of rows and not needed for our tracking goals, the entire column was dropped from our dataset memory: `df.drop(columns=['Middle Name'], inplace=True)`
- **Filling Empty Store Names:** To create clean filters for our dashboard, missing entries in the Store Name field were replaced with an 'Unknown' string flag: `df['Store Name'] = df['Store Name'].fillna('Unknown')`
- **Creating a Customer Type Flag:** We built a new column called Customer Type. It automatically checks if a store name exists, splitting our users into 18,484 'Individual' (B2C) buyers and 1,336 wholesale 'Store' (B2B) businesses: `df['customer type'] = df['Store Name'].apply(lambda x: 'Store' if x != 'Unknown' else 'Individual')`
- **Mapping Unclear Codes to Text:** Short codes ('IN', 'SC') in the Person Type column were mapped to human-readable names ('Individual', 'Store Contact'). The remaining missing rows were filled with 'Store Without contact' to keep things consistent: `df['Person Type'] = df['Person Type'].map({'IN': 'Individual', 'SC': 'Store Contact'}).fillna('Store Without contact')`
- **Final Polish & Export:** Any remaining empty text fields in customer first and last names were replaced with 'Unknown' and 'Customer' placeholders. The final data was verified to have zero empty rows and exported directly to dcustomer.csv.

6. Formatting Data in Power Query

The clean customer CSV files were imported into Power Query to link lookup variables and standardize database formats:

- **Consolidating Customer Fields:** We merged separate First Name and Last Name string columns into a single column called Full Name. This makes building customer charts and tables much easier.
- **Formatting Commission Fields:** To ensure proper numeric math in our sales charts, the Commission Pct column was transformed from a standard decimal format type into an explicit percentage data structure.
- **Enriching Territory Dimensions:** Regional database files were updated by grouping separate locations logically. We reordered columns to build clean filtering paths and mapped country tracking codes uniformly.

7. Data Model Design & Dashboard Layout

The clean datasets were organized into an analytical **Star Schema** inside Power BI. This setup puts transaction fact tables in the center and points surrounding lookup tables toward them using one-to-many filtering connections. This configuration keeps dashboard calculations fast and minimizes processing lags.

Data Model Fact Table Assignments

- **F_Sales:** Holds transaction entries used to calculate sales numbers, track pricing matrices, and monitor gross revenue.
- **F_Purchasing:** Serves as the tracking baseline for vendor reliability, recording items ordered, incoming delivery volumes, and arrival timelines.
- **F_Transaction_History:** Consolidates warehouse stock updates, product changes, and physical movements into a single log file.
- **F_Inventory_Snapshot:** Records physical product quantities across regional warehouse locations to monitor asset storage costs.

Completed Dashboard Layout Pages

- **HOME Page**
    - Acts as the landing page and main entry point for the business intelligence report.
    - Displays foundational project details, the team name, and corporate program info.

    - Outlines the technical data pipeline and lists specific team roles and project management responsibilities.

- **OVERVIEW Page**
    - Provides a high-level corporate summary of total enterprise operational health.
    - Highlights key performance indicators (KPIs) to monitor operational health and support executive decision-making.
    - Gives stakeholders a general view of company status to help maximize business profitability.

- **SALES Page**
    - Tracks gross profit margins, global revenue trends, and promotional campaign success rates
    - Identifies top-performing products and uncovers localized consumer purchasing patterns.
    - Segments sales metrics between individual consumers (B2C) and wholesale commercial businesses (B2B) using engineered type flags
- **INVENTORY Page**
    - Monitors physical stock counts and product distribution metrics across regional warehouse locations
    - Tracks structural item movement logs to highlight stock ins, stock outs, and snapshot tracking lines
    - Helps inventory managers optimize warehouse space utilization and prevent product stockouts
- **PURCHASES Page**
    - Displays comprehensive procurement transaction records, order volumes, and ordering dates
    - Monitors detailed line item financial details by tracking unit prices, applied discounts, and line totals.
    - Categorizes purchase order lifecycle histories into clear stages like pending, approved, or completed

- **VENDORS Page**
    - Evaluates active supplier profiles, names, and general partner behaviors

- Uses credit rating metrics to ensure third-party partners meet enterprise reliability benchmarks
- Highlights preferred vendor status evaluations to help organize long-term procurement planning

- **SUPPLY CHAIN Page**
  - Identifies fulfillment bottlenecks and supply chain roadblocks before they slow down business operations.
  - Evaluates product delivery quality by monitoring supplier rejection rates and item failure volumes.
  - Analyzes shipping method data and baseline handling costs to improve overall delivery logistics